Week 1 · Task 2 — Networking Basics for Defenders

1. The OSI 7-Layer Model Architecture

The Open Systems Interconnection (OSI) model standardizes the communication functions of a telecommunication or computing system into seven distinct conceptual layers. Data transitions downwards from Layer 7 to Layer 1 at the transmitter, and flows upwards from Layer 1 to Layer 7 at the receiving host.

Layer # & Name	Data Unit (PDU)	Primary Functions & Responsibilities	Protocols / Hardware
L7: Application	Data	Provides network services directly to end-user software applications. Interacts with software to implement communication functions.	HTTP, HTTPS, FTP, SMTP, DNS, SSH, Telnet
L6: Presentation	Data	Formats, translates, encrypts, and compresses data. Ensures data is in a readable format for the application layer.	SSL/TLS, JPEG, PNG, ASCII, MPEG, GIF
L5: Session	Data	Establishes, manages, maintains, and terminates communication sessions between local and remote applications.	NetBIOS, PPTP, RPC, Sockets, Session API
L4: Transport	Segment (TCP) / Datagram (UDP)	Provides end-to-end communication control, error detection, flow control, windowing, and multiplexing via port numbers.	TCP, UDP, SCTP
L3: Network	Packet	Handles logical addressing (IP addresses), path determination, packet forwarding, and routing across subnetworks.	IPv4, IPv6, ICMP, ARP, IPsec, Routers
L2: Data Link	Frame	Provides node-to-node data transfer, physical addressing (MAC), media access control (MAC sublayer), and error framing (LLC).	Ethernet (802.3), Wi-Fi (802.11), Switches, Bridges
L1: Physical	Bits	Transmits raw, unstructured bit streams over physical transmission mediums (electrical signals, optical pulses, radio waves).	Cables (Cat6, Fiber), Hubs, Repeaters, Network Cards

Key Concept — Encapsulation & Decapsulation:

As data moves down the stack, each layer adds its own control header (and trailer at Layer 2) to the payload received from the layer above: **Data → Segment/Datagram → Packet → Frame → Bits**. On the receiving end, the process reverses (Decapsulation), stripping headers at each respective layer.

2. Essential Network Protocols & Port Numbers

Port numbers operate at Layer 4 (Transport Layer) to map network traffic to specific system processes or application services. Standardized port numbers are categorized into Well-Known Ports (0–1023), Registered Ports (1024–49151), and Dynamic/Private Ports (49152–65535).

Port #	Protocol	Transport	Description & Usage	Security Implications
20, 21	**FTP**	TCP	File Transfer Protocol (Data & Control streams)	Unencrypted plaintext credentials
22	**SSH / SFTP**	TCP	Secure Shell / Secure File Transfer Protocol	Encrypted remote terminal access
23	**Telnet**	TCP	Unencrypted text communications / terminal access	Deprecated; highly insecure (plaintext)
25	**SMTP**	TCP	Simple Mail Transfer Protocol (Email routing)	Prone to relay abuse if unauthenticated
53	**DNS**	UDP / TCP	Domain Name System resolution	UDP for lookup, TCP for zone transfer
67, 68	**DHCP**	UDP	Dynamic Host Configuration Protocol (IP leasing)	Client port 68, Server port 67
80	**HTTP**	TCP	Hypertext Transfer Protocol (Web access)	Unencrypted web traffic
110	**POP3**	TCP	Post Office Protocol v3 (Retrieves email)	Plaintext; replaced by POP3S (Port 995)
123	**NTP**	UDP	Network Time Protocol (Clock synchronization)	Critical for log timestamps & auth
143	**IMAP**	TCP	Internet Message Access Protocol (Email sync)	Replaced by IMAPS (Port 993)
161, 162	**SNMP**	UDP	Simple Network Management Protocol	Community strings used for auth
389	**LDAP**	TCP / UDP	Lightweight Directory Access Protocol	Directory lookups; use LDAPS (636)
443	**HTTPS**	TCP	HTTP Secure (HTTP over TLS/SSL encryption)	Standard encrypted internet web traffic
445	**SMB**	TCP	Server Message Block (Windows file/print sharing)	High vulnerability target (e.g. WannaCry)
3389	**RDP**	TCP / UDP	Remote Desktop Protocol (Windows GUI access)	Target for brute-force

ransomware attacks

3. Domain Name System (DNS) Resolution Workflow

The Domain Name System (DNS) translates human-readable domain names (e.g., www.example.com) into machine-routable IP addresses (e.g., 93.184.216.34). The resolution process relies on a hierarchical tree structure of specialized servers.

- **Step 1: Local Cache Check:** The user requests a URL in their web browser. The operating system checks the local browser cache, OS resolver cache, and local 'hosts' file. If the IP address is found locally, the query ends immediately.
- **Step 2: Recursive DNS Resolver Query:** If uncached, the OS sends a recursive DNS query to the configured Recursive Resolver (typically provided by the ISP or public resolvers like 8.8.8.8 or 1.1.1.1). The recursive resolver handles the resolution chain on behalf of the client.
- **Step 3: Root Name Server Query:** The Recursive Resolver queries one of the 13 logical Root Servers ('.'). The Root Server does not store individual IP addresses; instead, it responds with a referral to the appropriate Top-Level Domain (TLD) Name Server based on the domain extension (e.g., .com, .org, .edu).
- **Step 4: TLD Name Server Query:** The Resolver queries the designated TLD Server (e.g., .com TLD server). The TLD server inspects the requested domain and returns the IP address of the Authoritative Name Server responsible for that specific domain.
- **Step 5: Authoritative Name Server Query:** The Resolver queries the Authoritative Name Server. Because this server holds the master DNS record (A / AAAA records), it returns the definitive destination IP address back to the Recursive Resolver.
- **Step 6: Result Caching & Client Response:** The Recursive Resolver caches the IP address locally for the duration defined by the record's Time-To-Live (TTL) value and returns the IP to the user's computer.
- **Step 7: HTTP Connection Assembly:** The client browser receives the IP address and initiates a TCP 3-Way Handshake (SYN, SYN-ACK, ACK) followed by a TLS handshake (if HTTPS) to establish a connection with the web server.

Key DNS Record Types

Record Type	Full Name	Description / Purpose
A	Address Record	Maps a domain name to an IPv4 address (32-bit).
AAAA	IPv6 Address Record	Maps a domain name to an IPv6 address (128-bit).
CNAME	Canonical Name	Aliases one domain name to another (domain redirection).
MX	Mail Exchanger	Directs incoming email to the domain's designated mail servers.
TXT	Text Record	Holds machine-readable data (used for SPF, DKIM, and domain verification).

Completed rooms | Created rooms | Skills matrix | Certificates | Badges | Yearly activity

DNS in Detail
Learn how DNS works and how it helps you access internet services.
Easy
Free | Walkthrough

What is Networking?
Begin learning the fundamentals of computer networking in this bite-sized and interactive module.
Info
Free | Walkthrough

Intro to LAN
Learn about some of the technologies and designs that power private networks
Info
Free | Walkthrough

Pentesting Fundamentals
Learn the important ethics and methodologies behind every pentest.
Easy
Free | Walkthrough

Security Principles
Learn about the security triad and common security models and principles.

Offensive Security Intro
Hack your first website (legally in a safe environment) and experience an ethical hacker's job.

Careers in Cyber
Discover the perks and your place within Cyber Security.

Task 1 Introduction

Task 2 OSI Model

Before we start, we should note that the OSI model might initially seem complicated. Don't worry if you encounter cryptic acronyms, as we provide examples of the OSI model layers. We assure you that by the time you finish this module, this task will feel like a piece of cake.

The OSI (Open Systems Interconnection) model is a conceptual model developed by the International Organization for Standardization (ISO) that describes how communications should occur in a computer network. In other words, the OSI model defines a framework for computer network communications. Although this model is theoretical, it is vital to learn and understand as it helps grasp networking concepts on a deeper level. The OSI model is composed of seven layers:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

The numbering starts with the physical layer being layer 1, while the top layer, the application layer, is layer 7. To help you remember the layers from bottom to top, you can use a mnemonic such as "Please Do Not Throw Spinach Pizza Away." You can check the Internet for other easy-to-remember acronyms if this helps you memorise them. Remembering the OSI model layers with their layer numbers is important; otherwise, you will struggle to understand terms such as "layer 3 switch" or "layer 7 firewall."